

THERESA STADLER

Data Privacy and Trustworthy AI

✉ theresa.stadler@epfl.ch

👤 Personal Web

🎓 GoogleScholar

🌐 LinkedIn

SUMMARY

Researcher in data privacy and trustworthy AI with experience across industry and academia

SKILLS

Programming: Python, R, jupyter, git

Topics: Privacy-preserving Machine Learning, Synthetic Data, Data Sharing

EXPERIENCE

Researcher & Lecturer - EPFL (CH) <i>Research and Teaching</i> Research on privacy-enhancing technologies, such as, synthetic data, privacy-preserving machine learning, and differential privacy. Lectures at BSc and MSc level on information security and privacy.	Since 2025
PhD Research Assistant - EPFL (CH) <i>Research and Teaching</i> Several high impact publications in the fields of data privacy and trustworthy machine learning.	2019 – 2024
Data Scientist - Privitar (UK) <i>Product Development, Research, and Consulting</i> Designed, developed, and prototyped enterprise software that implements privacy-enhancing technologies at scale. Several consulting projects for both public and private sector organisations.	2016 – 2019
Graduate Student Research Assistant - Werner Reichardt Centre for Integrative Neuroscience (DE) <i>Experimental Research and Data Analysis</i> Statistical models of visual information processing in retinal ganglion cells.	2015-2016
Student Research Assistant - University of Erlangen (DE) <i>Experimental Research and Data Analysis</i> Biophysical modelling of the molecular mechanisms of chronic pain disorders.	2012 - 2014

EDUCATION

PhD in Computer Science - EPFL (CH), SPRING Lab PhD Thesis: On the Fundamental Limits of Privacy-Enhancing Technologies.	2019 - 2024
MSc in Computational Neuroscience - University of Tübingen (DE) Lectures in Statistics, Machine Learning, Dynamic Systems, and Neuroscience	2014 - 2016
BSc in Biomathematics - University of Erlangen (DE) Lectures in Statistics, Linear Algebra, Physics, and Biology	2011 - 2014

GRANTS & AWARDS

Nominee for the EPFL Doctorate Award - EPFL (CH)	2024
Teaching Assistant Award - EPFL IC (CH)	2021
Graduate Grant - Studienstiftung des Deutschen Volkes (DE)	2011 - 2016

SELECTED INVITED TALKS

Talk	On the Fundamental Limits of Privacy-Enhancing Technologies - Nokia Bell Labs Invited talk at the Responsible AI seminar.	2025
Lecture	On the fundamental limits of privacy-preserving data sharing technologies - Brussels Privacy Hub Lecture at the Summer Academy For Global Privacy Law 2024	2024
Panel	Looking beyond the EU data strategy: Where next for data use and regulation? - CPDP Panel discussion on the future of data use and regulations	2023
Lecture	Synthetic data as a privacy mechanism - A cautionary tale - MIT Invited lecture in the Health Science and Technology Program	2022

SELECTED MEDIA COVERAGE

Podcast	Privacy Engineering - They Talk Tech Podcast Available at frauen-technik.podigee.io	2025
News Article	Warum wollen plötzlich alle Luca? - Eva Wolfangel, <i>Die Zeit</i> Available at zeit.de	2021
News Article	EU privacy experts push a decentralized approach to COVID-19 contacts tracing - <i>TechCrunch</i> Available at techcrunch.com	2020
News Article	Coronavirus apps: the risk of slipping into a surveillance state. - <i>Financial Times</i> Available at ft.com	2020

LANGUAGES

English - Fluent, **German** - Native, **French, Italian** - Good conversational skills

SELECTED PUBLICATIONS

- 2024**
- T. Stadler, B. Kulynych, N. Papernot, M. Gastpar, and C. Troncoso. The fundamental limits of least-privilege learning. In *Proceedings of the 41th International Conference on Machine Learning (ICML 24)*, 2024
- 2022**
- T. Stadler, B. Oprisanu, and C. Troncoso. Synthetic data – Anonymisation Groundhog Day. In *31st USENIX Security Symposium (USENIX Security 22)*, 2022
- T. Stadler and C. Troncoso. Why the search for a privacy-preserving data sharing mechanism is failing. *Nature Computational Science*, 2022
- C. Troncoso, T. Stadler, D. Bogdanov, E. Bugnion, S. Chatel, C. Cremers, S. Gürses, J.-P. Hubaux, D. Jackson, J. R. Larus, et al. Deploying decentralized, privacy-preserving proximity tracing. *Communications of the ACM*, 2022
- 2021**
- T. Stadler, W. Lueks, K. Kohls, and C. Troncoso. Preliminary analysis of potential harms in the luca tracing system. *arXiv preprint arXiv:2103.11958*, 2021
- 2020**
- C. Troncoso, M. Payer, J.-P. Hubaux, M. Salathé, J. Larus, E. Bugnion, W. Lueks, T. Stadler, A. Pyrgelis, D. Antonioli, et al. Decentralized privacy-preserving proximity tracing. *arXiv preprint arXiv:2005.12273*, 2020
- V. von Wyl, S. Bonhoeffer, E. Bugnion, M. A. Puhon, M. Salathé, T. Stadler, C. Troncoso, E. Vayena, and N. Low. A research agenda for digital proximity tracing apps. *Swiss Medical Weekly*, 2020
- M. Salathé, C. L. Althaus, N. Anderegg, D. Antonioli, T. Ballouz, E. Bugnion, S. Capkun, D. Jackson, S.-I. Kim, J. Larus, et al. Early evidence of effectiveness of digital contact tracing for sars-cov-2 in switzerland. *medRxiv*, 2020

PATENTS

- 2023**
- J. D. McFALL, C. C. Cabot, T. J. Moran, K. F. P. Guinamard, V. M. Eatwell, B. T. Pickering, P. D. Mellor, T. Stadler, A. Petre, C. A. Smith, et al. Computer-implemented privacy engineering system and method, Nov. 9 2023. US Patent App. 18/349,223
- 2022**
- C. C. Cabot, K. F. P. Guinamard, J. D. McFALL, P.-a. Maudis, P. Hector, B. T. Pickering, T. Stadler, J.-a. Tay, and S. Weller. Method or system for querying a sensitive dataset, Sept. 1 2022. US Patent App. 17/618,765